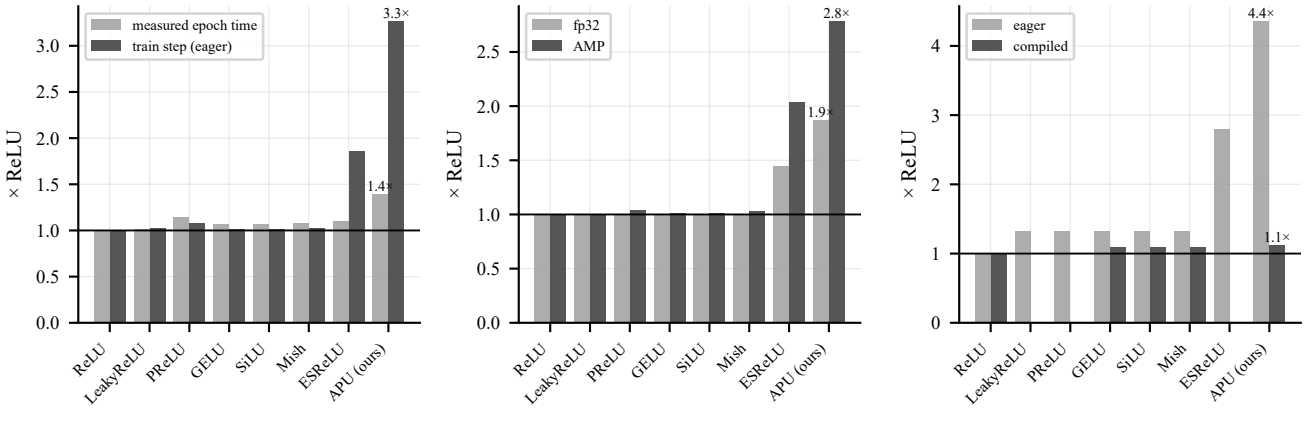
(a) training cost
measured epoch time
train step (eager)
× ReLU
3.3×
1.4×
(b) inference latency, batch 128
fp32
AMP
2.8×
1.9×
(c) peak training memory
eager
compiled
4.4×
1.1×
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)